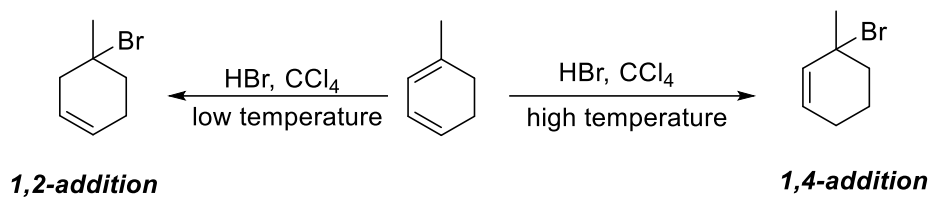
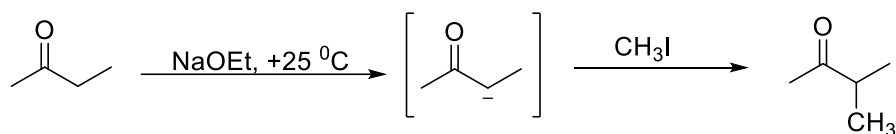


Q1. Write the major product for the following reactions. (1 x 2=2)

a)

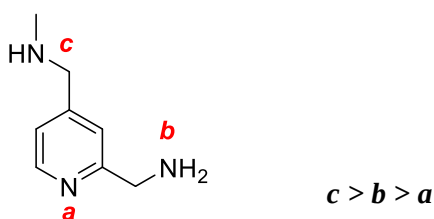


b)

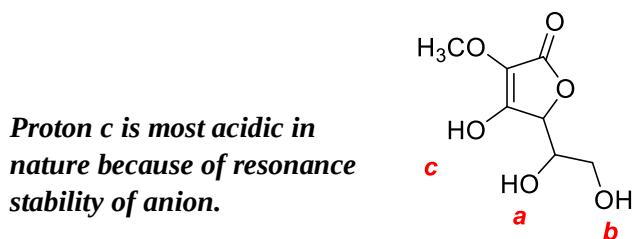


Q2. Answer the following questions. (1 x 2= 2)

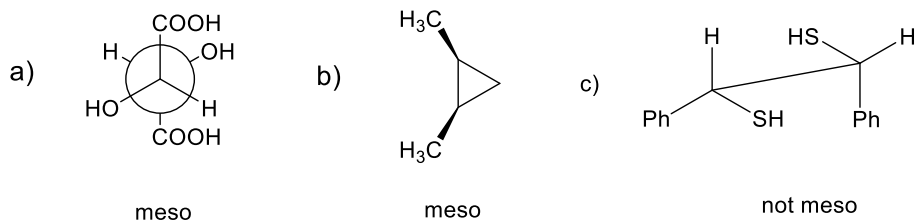
a) Arrange the following nitrogen in ascending order of their relative order of basicity?



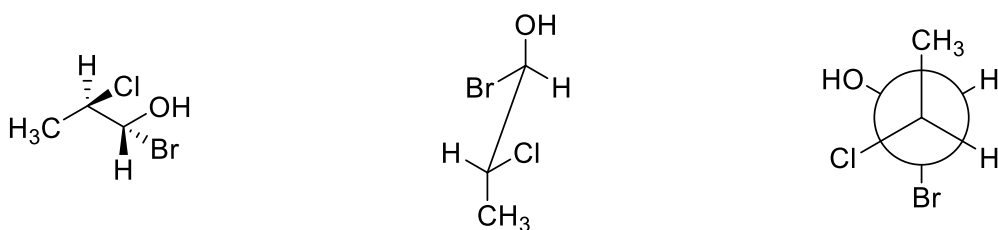
b) Among the following hydrogens which is more acidic in nature?



Q3. Which of the following is(are) meso compounds? (1)



Q4. Draw the most stable Newmann projection of the following compound. (2)



Q6. What is $[\alpha]$ for pure R if $[\alpha]$ for a mixture of enantiomer is -16° ? Given that the mixture is 70% R and 30% S? (1)

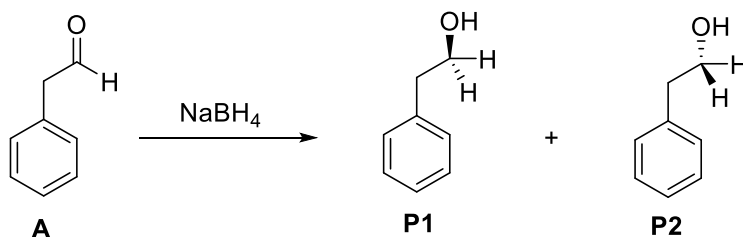
$$[\alpha]_{\text{mix}} / [\alpha]_{\text{pure}} \times 100 = \text{e.e.}$$

$70\% \text{ R} + 30\% \text{ S} = 40\% \text{ R}$

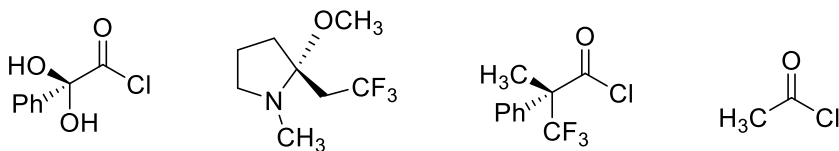
$$-16 / [\alpha]_{\text{pure}} \times 100 = 40$$

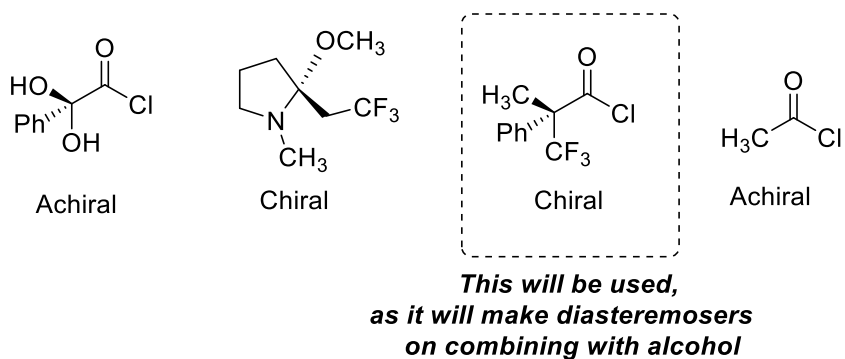
$$[\alpha]_{\text{pure}} = -40^\circ.$$

Q7. Dr. "X," an organic chemist, carried out the reduction of compound **A** using sodium borohydride (NaBH_4), resulting in a mixture of the enantiomers **P1** and **P2**.

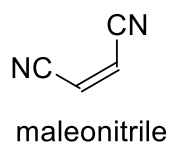
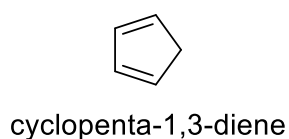


Dr. "X" intends to isolate the enantiopure compound **P2** but is uncertain about the appropriate procedure and reagent selection. Please provide guidance on selecting a suitable reagent from the options provided and outline the procedure schematically for obtaining the enantiopure compound. (2)

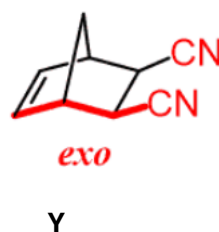
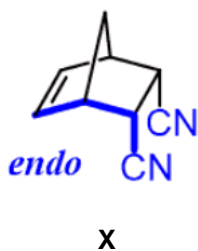




Q8. Cyclopentadiene on treatment with maleonitrile at room temperature gives a compound **X** which on heating at 120 °C changes to compound **Y**. Both compound **X** and **Y** have identical molecular formula. (2+2=4)

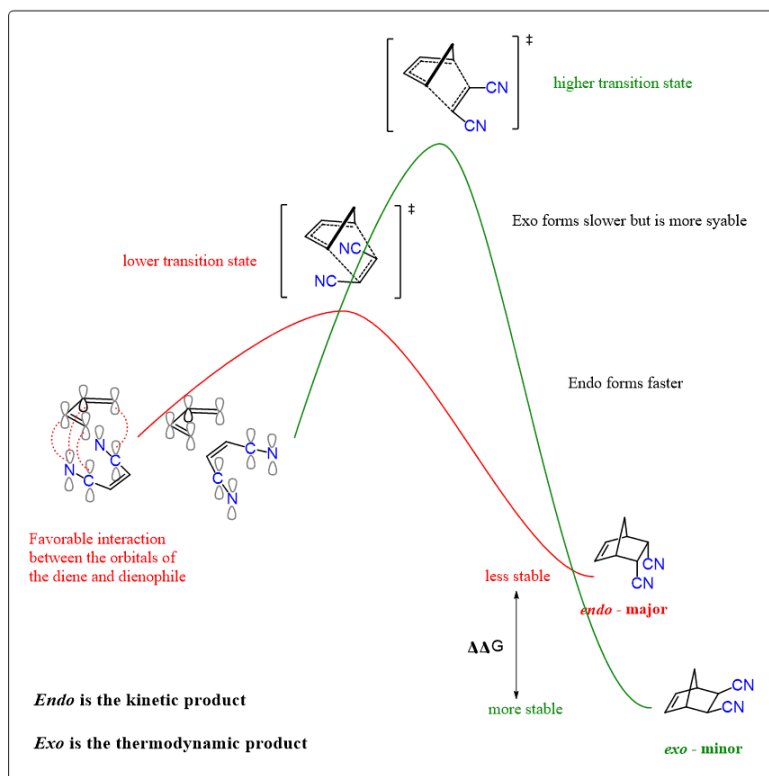


(a) Write down the structure of **X** and **Y**.

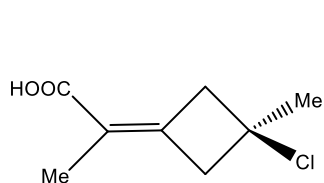


(b) Draw the energy profile diagram for the formation of **X** and **Y** on the same graph along with the structure of transition states.

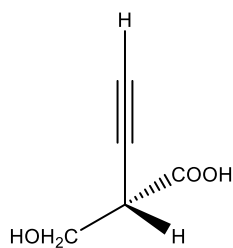
Energy diagram of the formation of *endo* and *exo* products in the Diels-Alder reaction



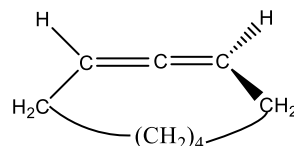
Q9. Assign the absolute configuration to the compounds given below. (2)



S

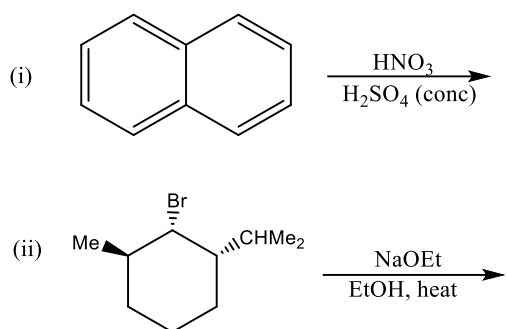


S



R

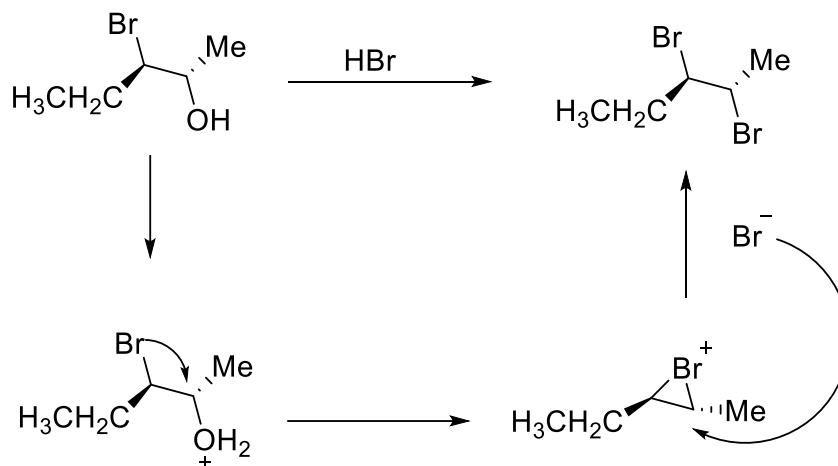
Q10. Which of the following reactions will show kinetic hydrogen isotope effect? Write the product formed and give a suitable reason justifying your answer. (3 marks)



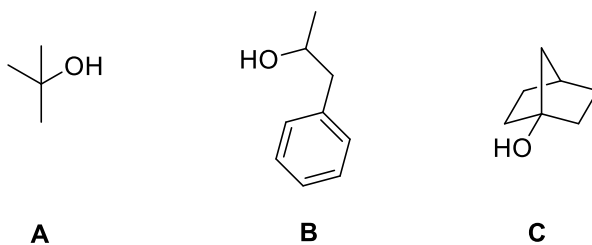
(ii) will show *KHIE* as it involves the *C-H* bond breaking at first step which is the rate determining step.

Q11

(a) Explain the following observation with suitable mechanism. (1)

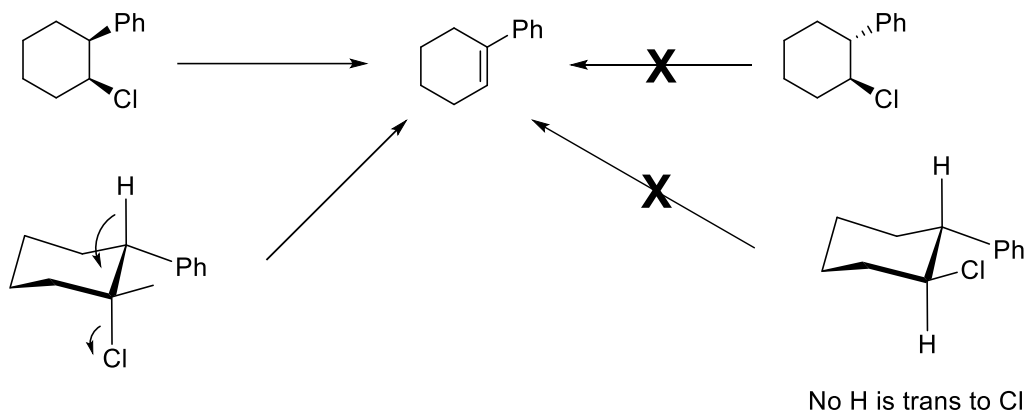


(b) Explain which among the following do not undergo an **E1** elimination with strong acid? (1)



Compound C will not undergo an *E1* elimination with strong acid because for an *E1* mechanism it has to form a carbocation, and a carbocation at bridge head is not possible.

Q12. (a) Cis-2-phenylcyclohexyl chloride reacts with potassium tert-butoxide in tert-butyl alcohol at 50 °C to give 1-phenylcyclohexene. Trans-2-phenylcyclohexyl chloride does not give any alkene under the same conditions. Please explain the observation. (2)



Q13: Fill in the Blank (0.5*6)

- (i) At 300 K, the free energy of $CO_2(g)$ at 1 atm is **smaller** than $CO_2(g)$ at 5 atm.
(greater/smaller)
- (ii) At 1 atm and 300 K, the free energy of $CO_2(g)$ is **smaller** than $CO_2(l)$. (greater/smaller)
- (iii) The melting temperature of ice will **decrease** with increase in pressure
(increase/decrease)
- (iv) The boiling point of 0.1 M aqueous NaCl solution will be **greater** than 100 °C.
(greater/lower)
- (v) For the reaction, $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, $\Delta H = -120 \text{ kJ mol}^{-1}$. The equilibrium constant of this reaction at 300 K will be **greater** than at 400 K (greater/smaller)
- (vi) The equilibrium constant for a reaction $A(g) + 2B(g) \rightleftharpoons 2C(g) + D(g)$ is 64. If 1 mol of A is mixed with 2 mol of B in a closed container, the number of mol of $A(g)$ at equilibrium will be **0.2**

Q14: Fill in the Blank:

A standard sodium carbonate solution is prepared by dissolving 10.6 g of the crystals in 250 mL water. 25 mL of this solution neutralises 10 mL of a HCl solution. (3)

- (i) The concentration of sodium carbonate solution is **0.4 M**
- (ii) The number of millimole of sodium carbonate in 25 mL of sodium carbonate solution is **10**
- (iii) The molarity of HCl solution is **2 M**